

Empirical Analysis of Transformers on Graph Connectivity

Part IX: Beyond the Diameter — Out-of-Distribution Path Generalization and the Role of Endpoints

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Abstract

Report VIII showed a standard connectivity transformer, trained on unions of disjoint paths and evaluated with a similarity read-out, resolving an unbalanced two-component split far beyond the doubled distance wall $2 \cdot 3^L$, and collapsing to “everything connected” on a structurally different two-component graph (two cycles) that the same training distribution never contains. Both results are measured *inside* a training distribution that already covers every two-segment split of the canvas, so neither one distinguishes a genuinely general mechanism (the two open *endpoints* of a path) from plain in-distribution coverage. This report is a plan for two threads that separate the two readings: (A) a search for a case of path learning that generalises to structure the model was *never* trained on, and (B) the same systematic training regime applied to *cycles* (which have no endpoints at all), to test whether endpoints are truly load-bearing or whether cycles, trained properly, are simply another counterexample to “the diameter is the difficulty measure.” No experiment in this report has been run yet; this document states the questions, the established background, and the planned setups.

1 Introduction

1.1 What Reports I–VIII established

We treat the following as established (documented, multi-seed, in Reports I–VIII); the new work in this report must stay consistent with them.

- **There is a sharp, architectural distance wall at 3^L** ($= 9$ at $L = 2$), which moves with depth and appears even on graph families seen $\sim 10^8$ times in training (Reports III–IV).
- **The model computes distance-bounded matrix powering, not a bounded traversal**, and a soft, learned “node budget” on dense structure is a removable data prior, not a second capacity (Report V).
- **Parallel routes and repeated local structure both help, within capacity, in data-shaped ways** rather than architecturally for free (Report VI).
- **A similarity read-out** ($z_{ij} = \text{scale} \cdot \cos(h_i, h_j) + \text{bias}$) **doubles the single-route reach to $2 \cdot 3^L$** and makes behaviour far more robust to the trunk’s imperfect, graded cross-component mixing than the linear read-out (Report IV, confirmed mechanistically in Report VII).
- **An unbalanced two-component split resolves far beyond the distance wall; a balanced split does not, with the same weights** (Report VI, Thread B) — opening the trunk (Report VII) showed why: on a graph with exactly two components, once one is small enough to resolve completely, the model gets an extra, graded completion signal for the *other* component, built by the same two-layer attention machinery that does ordinary beyond-capacity reach, not a separate circuit.

- **Report VIII collected the full mechanistic battery under the now-standard similarity read-out** (behavioural sweep, read-in geometry, attention scores, the exact node-to-node contribution, a new causal edge-ablation contribution, layer ablation, the three-component falsification test) on the disjoint-paths-trained, $n=40$ checkpoints, and added a stagewise probe locating *where* in the forward pass the completion signal is assembled: attention layer 2, not either feed-forward step, builds most of the far-reach; MLP 2’s distinctive job is sharpening the *cut*.
- **Report VIII also tested two *cycles* in place of two chains**, on the same disjoint-paths-trained checkpoints (which never saw a cycle in training): the model collapsed to predicting the entire graph connected at *every* split and every seed (predicted-positive rate exactly 1.000, cut exactly 0.000) — read as a generic out-of-distribution collapse on an unfamiliar canvas, not a clean test of whether path endpoints specifically matter, since the training distribution never contained a cycle of any kind.
- **The training distribution behind every two-component result since Report VI** is `path_union`-style: an online stream of unions of 1–4 disjoint paths that, together, partition all n canvas nodes. Crucially, this stream already contains *every* two-segment split of the canvas as a special case (Report VII, §sec:intro there) — so any result showing the model resolving a particular two-component split is, by construction, an in-distribution result, not evidence of generalisation to unseen structure.

1.2 The new question: is the diameter always the right measure, and are path endpoints load-bearing?

Report VIII’s two-cycles result and the underlying two-chain completion signal (Report VI/VII) together suggest that *diameter alone* is not always the right notion of difficulty: a model can resolve pairs far beyond $2 \cdot 3^L$ on one graph shape and fail completely on a same-size, same-split graph of a different shape. But every demonstration of “beyond the wall” so far sits *inside* a training distribution that already covers the test structure. This report asks two questions in sequence, each stated as a plan of experiments to be run in a later session, not results already in hand.

1. **Is there a case of path learning that is genuinely out-of-distribution?** All the evidence so far only shows that, *in distribution*, the model learns beyond the doubled wall by using the two open endpoints of a path. Is that a general skill that transfers to structure never seen in training (a different diameter cutoff, a different number of components, a different internal density), or is it tied to having seen exactly this kind of structure during training?
2. **Are the two open endpoints of a path truly load-bearing, or is the two-cycles failure of Report VIII simply an artefact of never having trained on a cycle at all?** Repeating the same systematic training regime used for paths, but on cycles, and reading the identical mechanistic battery, would settle whether cycles are architecturally unlearnable in this respect or whether they are simply unfamiliar — in which case cycles become a *second* counterexample to “the diameter is the difficulty measure,” on equal footing with the path result, rather than evidence that endpoints are special.

1.3 Two candidate readings, stated before running anything

As in every previous report, we state the candidate readings up front so the planned experiments are genuine tests, not a search for evidence of an assumed answer.

- **Reading 1 — endpoints are load-bearing.** The completion signal is tied specifically to the existence of an open, degree-1 endpoint; a structure without one (a cycle) cannot

support it no matter how it is trained, and a path structure stripped of its usual training coverage (e.g. capped diameter, fewer path counts) loses the skill on genuinely unseen splits.

- **Reading 2 — endpoints are incidental; coverage/familiarity is what matters.** Given a training distribution that systematically exposes the model to a given two-component shape (paths *or* cycles) across a range of splits, the model acquires an analogous completion signal regardless of whether the shape has open endpoints; the two-cycles failure of Report VIII is then simply because cycles were never trained on, not because they lack endpoints.

Neither reading is assumed; the point of Threads A and B below (§3) is to produce evidence that could support either one.

2 Setup

Model. The same base model as Reports III–VIII: the RoBERTa-faithful standard transformer, $L = 2$, single head, $d_{\text{model}} = 512$, normalised-ReLU attention. As in Report VIII, we standardise on the **similarity** read-out ($z_{ij} = \text{scale} \cdot \cos(h_i, h_j) + \text{bias}$) throughout this report, for the same reason given there: it makes behaviour far more robust to the trunk’s imperfect, graded cross-component mixing than the linear read-out, without removing the underlying mechanistic signal (Report VII, §sec:res-similarity).

Canvas size: $n = 46$. Every new training run in this report uses $n = 46$ nodes, not the $n = 40$ used throughout Reports VI–VIII. Everything else about the training setup is kept identical to the disjoint-paths condition of Report VIII (same online distribution family where applicable, same number of training steps/samples, same architecture, same read-out). This is an explicit, non-negotiable instruction (§3 restates it per thread where relevant); the training/evaluation code from Reports VI–VIII is already parametric in the canvas size and is reused unchanged, not rewritten.

Test graphs. Each planned experiment below (§3) specifies its own test construction (a capped-diameter path union, a two-chain graph with an edge removed to create a third component, unions of 5–7 disjoint paths, a chain-plus-clique graph, a two-component cycle graph, and others). No test graphs have been generated yet; the exact generators to reuse or add are noted in §3 as implementation notes, to be resolved when each experiment is actually built.

3 Plan of experiments

This report is organised around two threads, in the order the open question in §1.2 states them. **No experiment below has been run.** Every subsection states a question, a planned setup, and stops at a “[*Data pending — experiment not yet implemented*]” marker; nothing here should be read as a result.

3.1 Preliminary check — does canvas size $n = 46$ reproduce the effect at all?

Before committing every training run in this report to $n = 46$ (§2), we first check, on two seeds, that the disjoint-paths/similarity condition at $n = 46$ reproduces the same endpoint-completion signature already established at $n = 40$ (Report VI/VII): a small split should resolve the long component far past the distance wall, a balanced split should not, with the same weights. Two independent training runs (seeds 1000 and 2000), identical in every respect to the Report VIII disjoint-paths condition except the canvas size, followed by the same split-sweep evaluation used

throughout Reports VI–VIII. *[Data pending — training launched by the user, not yet returned; code prepared and reused unchanged from Reports VI–VIII.]*

3.2 Thread A — out-of-distribution generalisation for path-trained models

Thread A is the more important of the two (explicitly, in the request behind this report): find at least one case where a path-trained model’s completion signal is tested on structure it provably never saw, rather than on a special case already covered by its training stream.

A.1 — Train with a capped diameter, test beyond it. Train on a union-of-paths distribution restricted to a maximum diameter (e.g. two-chain splits that never exceed diameter 18, the doubled wall $2 \cdot 3^L$), then evaluate on two-chain splits whose diameter *exceeds* that cap — structure the training stream never produced by construction. This is the most direct test of whether the completion signal generalises past what the model has been shown, or whether it is bounded exactly by what was covered in training. *[Data pending — not yet implemented.]*

A.2 — Less variety in the training distribution. The current disjoint-paths stream (unions of 1–4 path components) may be more varied than necessary for the model to acquire the endpoint-completion skill. Train instead on a narrower distribution (e.g. only two-chain splits, never 3 or 4 components) and test whether the completion signal still appears. If it does, the phenomenon does not depend on the richness of the training distribution (a more general, more interesting result); if it does not, the phenomenon is specific to the richer stream (a narrower but still reportable result). *[Data pending — not yet implemented.]*

A.3 — Genuinely unseen three-component test. Train *only* on two-chain splits (never three or more components), then evaluate on a graph built by removing one internal edge from a two-chain test graph, producing three components that were never a possibility under this training distribution (unlike the three-component test already run in Reports VII–VIII, whose training distribution could already produce three or four components). *[Data pending — not yet implemented.]*

A.4 — Existing checkpoints on 5/6/7-component unions. Using the already-trained `path_union` checkpoints (which only ever see 1–4 components in training), evaluate on unions of 5, 6, and 7 disjoint paths — component counts never produced during training — to see whether the endpoint-completion skill generalises to a number of components beyond what was covered. One long component plus $K-1$ short ones, with the short-component size chosen so the long component’s internal diameter exceeds 18 (the doubled wall), so a failure cannot be attributed to the ordinary distance wall alone. This experiment is eval-only on existing $n=40$ checkpoints and is the one exception to the $n=46$ rule of §2 (no retraining involved). *[Data pending — eval code implemented and smoke-tested, launched by the user, not yet returned.]*

A.5 — Two-chain-trained checkpoints on targeted structures. Train (four seeds, as throughout this project) on a range of two-chain splits only, confirm the endpoint-completion signal reappears, then evaluate the resulting checkpoints on structures chosen to isolate what actually matters: (i) a graph made of one chain and one clique (two components, only one of them with a genuine open endpoint); (ii) a two-component graph shaped like a two-chain split but where the *internal* (non-endpoint) nodes of each component carry additional edges among themselves, so the two components stay separate and still have clearly identifiable endpoints but are internally denser. The goal is to separate “needs the exact path shape” from “needs two components with recognisable endpoints.” *[Data pending — not yet implemented.]*

3.3 Thread B — are endpoints truly necessary? The analogous experiment on cycles

Thread B asks the mirror question directly: instead of treating the two-cycles collapse of Report VIII as evidence that endpoints are necessary, train the model on cycles the same systematic way paths were trained, and see whether an analogous completion signal appears.

B.1 — The same training regime as paths, applied to cycles. Repeat, with cycles in place of paths, the same kind of systematic training distribution used for the path results of Reports VI–VIII (an explicit, named distribution over cycle unions/splits, not an opaque mixture — consistent with the data principle introduced in Report VI, §13.2 of the project instructions), sweeping the split the same way, and evaluate *in-distribution* (the same kind of evaluation the path results received, as opposed to the purely out-of-distribution two-cycles test of Report VIII, where the training stream never contained any cycle at all). The aim is to see whether, and how, the model learns to handle cycles when they are not a foreign structure. *[Data pending — not yet implemented.]*

B.2 — The full Report VIII mechanistic battery, repeated on cycle-trained checkpoints. On the resulting cycle-trained checkpoints, repeat the entire battery assembled in Report VIII: the general behavioural results, the read-in geometry analysis, the attention scores, the exact node-to-node contribution, the layer ablation, a three-cycles falsification test (the structural analogue of the three-component split test), a two-cycles split sweep, and the layer-wise cosine geometry (stagewise probe). The purpose is to determine, before concluding that path endpoints are important or that cycles are intrinsically harder than other graph shapes, whether cycles — trained on properly — can *also* serve as a counterexample to “the diameter is the difficulty measure,” on the same footing as the path result. *[Data pending — not yet implemented.]*

4 Status

Most of §3 is still unimplemented. Two exceptions, both code-complete and smoke-tested but with no real data back yet: the preliminary $n=46$ check (§3.1, two training runs) and Thread A.4 (§3.2, an eval-only test on existing $n=40$ checkpoints). Everything else in this document exists to fix the questions, the established background, and the exact planned setups before any further code is written, so that the next session can proceed thread by thread — one experiment at a time, each smoke-tested on a toy checkpoint before touching real weights, with every number in a future revision of this report pulled from real data on real checkpoints, never written from memory or from partial output.